

URA Visiting Scholars Program Project Proposal

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Abstract

I describe the two activities I aim to complete with the support of the URA Visiting Scholars Program as a graduate student working on the Compact Muon Solenoid (CMS) experiment. Modernization of analysis pipelines and the implementation of new tools in analyses are essential to maximizing the physics potential of the data high energy physicists work with. In this project, I propose the implementation of a Parametrized Neural Network in the search for Beyond Standard Model Physics using the Standard Model Effective Field Theory. In the same spirit, I also detail the steps needed to develop a Coffea-compatible data discovery tool seamlessly integrates the early stage of analysis building into the rest of the analysis pipeline. Since my collaborators and relevant subject experts for these projects are located at Fermilab, being there allows me to complete these complementary projects as efficiently as possible.

1 Implementing a Neural Network in the search for Beyond Standard Model Physics using the Standard Model Effective Field Theory

1.1 Overview and Motivation

While the Standard Model (SM) of particle physics is highly successful at low energy scales, it is still an incomplete theory. The discovery of new particles would prove the existence of physics Beyond the Standard Model (BSM). Since such particles have not been observed directly yet, there is strong motivation for indirect searches looking for deviations from SM theoretical predictions.

The SM Effective Field Theory (SMEFT) is a powerful, model-agnostic framework for parameterizing potential deviations of experimental observations from SM predictions [1]. The SM Lagrangian, denoted by $\mathcal{L}_{SM}^{(4)}$, is treated as the lowest order term in an expansion, with higher order terms suppressed by Λ , which represents the energy scale of new physics [2].

$$\mathcal{L} = \mathcal{L}_{SM}^{(4)} + \frac{1}{\Lambda} \sum_k c_k^{(5)} Q_k^{(5)} + \frac{1}{\Lambda^2} \sum_k c_k^{(6)} Q_k^{(6)} + \mathcal{O}\left(\frac{1}{\Lambda^3}\right)$$

Here, $Q_k^{(d)}$ are effective field theory (EFT) operators of mass dimension d , and $c_k^{(d)}$ are Wilson Coefficients (WCs) that parametrize the interaction strength of the relevant SMEFT operator. Ignoring odd-dimensional operators, which violate lepton or baryon number, dimension-6 operators are expected to give the leading BSM contribution to relevant processes. Setting tighter bounds on the dimension-6 WCs allows the exploration of specific off-shell effects that can indicate and narrow down the possibility BSM phenomena.

The top quark sector is a promising area of interest for exploring new physics through the SMEFT framework. The Large Hadron Collider at CERN is a top-quark ‘factory’, which makes to quarks an ideal candidate for

study. The relevant final states are also easy to trigger on. The top quark is the heaviest particle in the SM, with several possible decay channels [3]. The large multiplicity of final states increases the scope of this study.

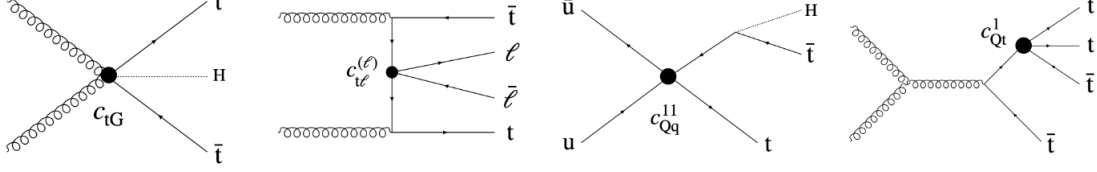


Figure 1: Example Feynman diagrams illustrating top-quark processes with vertices associated with the Wilson Coefficients c_{tG} , $c_{t\ell}^{(l)}$, c_{Qq}^{11} , c_{Qt}^1

1.2 The Case for Parametrized Neural Networks

There can be multiple WCs contributing to the kinematic properties seen in a single final state. The ability to narrow down the WCs responsible for certain kinematic effects will allow the setting of tighter, more accurate bounds on their values. The correlated nature of WCs makes it difficult to isolate effects observed due to a single WC. With analyses looking at some 26-odd WCs [3], the problem of correlated effects becomes combinatorially challenging.

Neural networks are already a well-established tool in high energy physics, from final state categorization to object reconstruction and triggering. These applications, however, involve an isolated problem without multiple parameters. Isolating and quantifying the kinematic effects of a highly correlated set of WCs, however, is a more complicated problem. A Parametrized Neural Network (pNN) is well-suited to such problems involving a range of closely related tasks.

Given a task with event-level features denoted by $\bar{x}$, and parameters $\bar{\theta}$, a pNN is fed an input vector of $\bar{x}$. After training, it outputs a function of $\bar{x}$ parametrized in θ . This is unlike a regular neural network, which does not allow the output to vary smoothly with a parameter of interest [4].

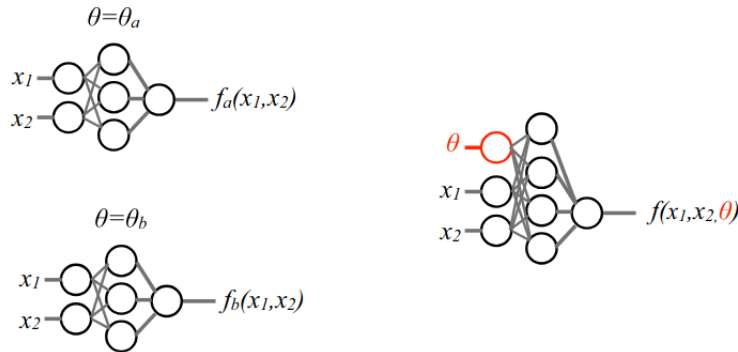


Figure 2: A schematic illustrating the difference between a regular neural network and a pNN. On the left, individual networks with input features (x_1, x_2) , trained with examples with a single value of some parameter $\theta = \theta_a, \theta_b$. The individual networks are purely functions of the input features. On the right, a single network trained with input features (x_1, x_2) as well as an input parameter θ ; such a network is trained with examples at several values of the parameter θ so the output can vary smoothly. [4]

Currently, analyses studying BSM effects in top-quark events using SMEFT categorize events according to the final state particle content. However, by selecting kinematic properties of interest as input variables, and WCs as parameters, it is possible to train a pNN on Monte Carlo simulations of processes of interest. The output will quantify the effects of different WCs on kinematic effects seen in different final states. This output can be used to devise a new categorization of events on the basis of WCs of interest instead of final state particle content.

While such an implementation of pNNs for the purpose of event selection and categorization has not been used in SMEFT studies, it is not unprecedented in other high energy physics analyses. For example, pNNs have been used for event categorization in searches for long-lived neutrinos at the Large Hadron Collider [5]. Kinematic variables of interest were the input, and the signal mass, m_N , was the single parameter used. A pNN outperforms a regular neural network used for similar purposes at points it was not trained on. Implementing a similar pNN-guided event categorization in SMEFT studies is thus an attractive prospect, with great potential to tighten the existing bounds on WCs.

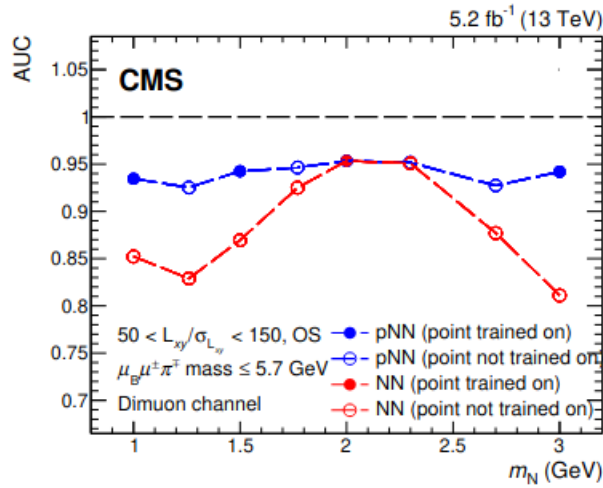


Figure 3: Validation of the use of a pNN for intermediate m_N mass points, for events in the $50 < L_{xy}/\sigma_{L_{xy}} < 150$, OS and low $l_B l^\pm \pi^\pm$ mass category in the dimuon channel: a pNN trained on masses $m_N = 1.0, 1.5, 2.0$, and 3.0 GeV (blue) and a NN trained on mass $m_N = 2$ GeV (red). The performance is shown by the AUC (Area Under the ROC Curve) curve, where a value at unity corresponds to a perfect separation between signal and background. All the points have $c\tau_N = 10$ mm. The full circles correspond to mass points on which the pNN and NN were trained on, while the open circles show mass points that have not been trained on. Note that the pNN performs well even for points that it was not trained on.[5]

1.3 Proposed Activity at Fermilab

Implementing a pNN to improve event categorization aligns with my goal of modernizing the analysis to make use of technology that maximizes physics insight.

Within the coming year, I aim to continue working upon training and implementing a pNN to create an alternative event categorization for improved bounds on WCs. My immediate goal is to create a proof-of-concept implementation of a pNN using a few weakly bound WCs and kinematic variables of interest. This involves selecting and finding out which kinematic characteristics are the best inputs, and training the pNN on Monte Carlo simulations. The performance of the pNN will be tested for improvements in bounds on the WCs of interest. Once the simplified implementation is satisfactory, I aim to extend the implementation to include all 26 WCs of interest in the analysis. This work will continue beyond the duration of the URA award.

In the past, I have worked on migrating the SMEFT analysis of multilepton top processes to the Coffea 2024 analysis framework hosted at the Coffea-casa analysis facility hosted at the University of Nebraska-Lincoln (see 2.2). This enabled the group to conduct the analysis using the more modern Coffea 2024 tools as opposed to command-line interfaces with longer times to insight. This project aligns with my goal of modernizing the analysis and implementing new tools to maximize its physics potential.

The large community of scientists involved in SMEFT research at the Fermilab LHC Physics Center is a great motivator for conducting this project there. I also look forward to attending the annual Effective Field Theory workshop at the Fermilab LHC Physics Center during my time there. It opens possibilities for training similar neural nets to be useful in a range of adjacent SMEFT analyses apart from the top-multilepton project. In addition to the presence of my collaborators and Fermilab sponsor, these are powerful motivators for me to conduct my work at Fermilab.

2 Building Coffea-compatible tool for CMS Dataset Discovery

2.1 Overview and Motivation

Maintaining 7 years' worth of Compact Muon Solenoid (CMS) datasets, along with their associated object corrections from Run 2 and Run 3, has become quite challenging. Powerful tools are necessary to collect and analyze the increasing number of eras of data and simulation processing. Additionally, the High-Luminosity upgrades to the Large Hadron Collider (HL-LHC) are set to be functional by 2030. It is projected to deliver an integrated luminosity of up to 4000 fb^{-1} [6]. The higher beam intensities of the HL-LHC require updated detectors, which in turn need higher trigger rates to maintain the same physics performance. The current computing infrastructure is inadequate to handle the resulting increased data readout.

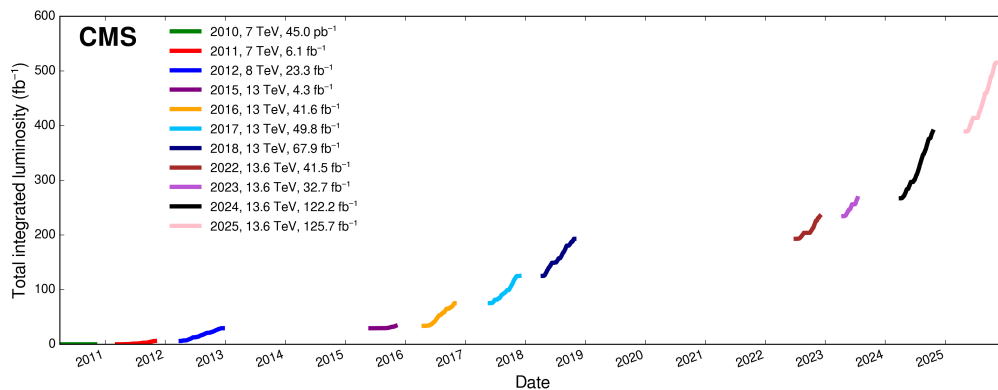


Figure 4: Graph showing the integrated luminosity delivered by the CMS experiment at the Large Hadron Collider. The amount of data collected has been ballooning over the years [7].

A standardized, scalable, end-to-end analysis pipeline is crucial for timely analysis, efficient use of computing resources, and ease of collaboration. The Institute for Research and Innovation in Software for High Energy Physics (IRIS-HEP) works on closing the remaining software and computing gaps to deliver HL-LHC science [8].

In order to realize the desired physics potential of the HL-LHC, new techniques and tools are essential for analysts. The increased challenges in data management and organization cannot be achieved simply by acquiring more disks and servers. It is necessary to leverage improved data delivery services and machine learning training and inference environments. New tools are actively being developed to address these

challenges. Additionally, these tools need to be made compatible with each other for a seamless analysis pipeline for end-to-end analysis optimization including systematics on a realistically sized HL-LHC (about 200 TB) end-user analysis dataset. This is addressed by the Analysis Grand Challenge [9] of the IRIS-HEP.

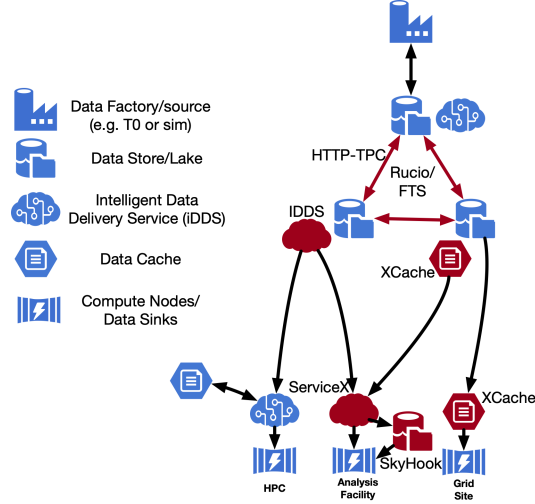


Figure 5: Schematic showing an overview of the conceptual components of the data management ecosystem for high energy physics. The technologies that the IRIS-HEP team contributes to are highlighted in red.

2.2 Existing Tools in the Analysis Pipeline

Coffea-Casa provides an environment for increasingly complex end-user analyses. Its columnar framework and user-friendly Jupyterlab interface make it a feature-rich, scalable, powerful analysis facility. Use of array programming and Python allows physicists to use more modern techniques and libraries, including but not limited to NumPy, SciPy, Pandas, Tensorflow and PyTorch [10].

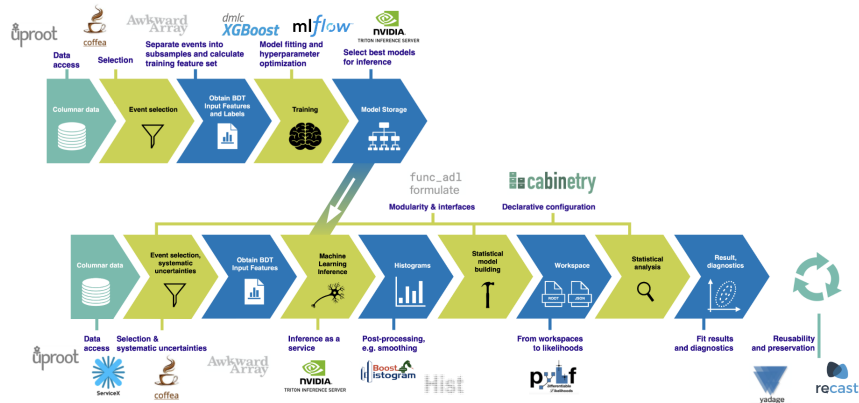


Figure 6: Representation of the Analysis Grand Challenge analysis pipeline with the involved tools and services [9].

In order to completely overhaul the old workflow of writing a C++ processor using the ROOT framework, several other tools are necessary. The awkward-array library deals with the complexity of high energy physics data, regularizing jagged arrays. Uproot allows reading and writing of ROOT files in pure Python and Numpy. ServiceX allows 'on-demand' columnar data delivery functionalities. These are all seamlessly integrated into the analysis pipeline[11].

However, the early steps of analysis building still hold room for improvement. Existing Coffea dataset tools are not yet tailored for common analysis use cases. Searching for and collating datasets or Monte Carlo samples in the early stages of an analysis remains a disjointed and unoptimized task. Multiple tools such as data management systems and metadata query services involved, making this step slow and confusing[12][13]. Data exploration is not possible in this framework, since it is not possible to query datasets by category or specific use case easily.

2.3 Proposed Activity at Fermilab

This part of the project aims to build a Jupyterlab plugin accessible within Coffea-Casa to streamline and automate the current dataset discovery and collation workflow. The desired features include Coffea and RDataFrame compatibility, and integrated workflow, and a graphical user interface for user-friendliness.

The user interface must be simple and intuitive. Once the appropriate databases are queried at each step in the workflow, the output plugin will be ready for use in a seamless analysis pipeline, whether in a Coffea processor, or an RDataFrame input.

I have worked on benchmarking and testing tools for the IRIS-HEP Analysis Grand Challenge for over 3 years. This project will be in some ways a continuation of my contributions to the Analysis Grand Challenge. Over the duration of the URA program, I will collect and study representative analysis use cases to automate and choose trigger streams and eras of data to include. These, along with suitable simulation samples, will form the options that users will see in the user interface. Consulting with the Coffea and data management experts at Fermilab will greatly help this project. Additionally, interacting with my peers at the LHC Physics Center at Fermilab allows me to conduct a focus group study that will help me understand the user options and functionalities that are most desired in a tool of this kind.

Summary

Implementing a Parametrized Neural Network in a SMEFT analysis modernizes the analysis and drives forward the search for BSM physics. The development of a Coffea-compatible data discovery tool seamlessly integrates the early stage of analysis building into the rest of the analysis pipeline. The URA Visiting Scholars Program will allow me to maximally contribute to both these related aims. Given the overlap in my collaborators on these two projects, and the fact that they are located at Fermilab, it is logical for me to work on these projects there as well. Additionally, the computing facilities at the LHC Physics Center enable easy testing and benchmarking. Given my interests in computing, this combined project provides a unique learning opportunity and allows me to contribute to two key areas in the same time period.

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